f(x+1) + f(x−1) = √2 f(x)

Animath — Stage olympique de Sophia-Antipolis · 2006 · France

---

Let $f : \mathbb{R} \to \mathbb{R}$ satisfy

$$f(x+1) + f(x-1) = \sqrt{2} f(x)$$

for every x in $\mathbb{R}$. Show that f is periodic.

What happens if $\sqrt{2}$ is replaced by the positive solution of $t^3 = t^2 + t + 1$?